

HAT-P-36b

R-0853

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_240404 · created 2026-07-28T20:56:25+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460404.6914 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.023
Transit depth (Rp/Rs)^2	1.96 +/- 0.65 [%]
Orbital Inclination (inc)	89.61 +/- 2.54
Airmass coefficient 1 (a1)	1.031 +/- 0.0068
Airmass coefficient 2 (a2)	-0.021 +/- 0.021
Scatter in the residuals of the lightcurve fit is	0.66 %
Best Comparison Star	#3 - [49, 73]
Optimal Aperture	3.55
Optimal Annulus	13.98
Transit Duration (day)	0.0729 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.14 $\pm$ 0.023 vs 0.126 (deviation 0.61 σ)
Duration fit vs published	0.0729 d vs 0.0925 d (deviation 4.67 σ)
Mid-time O-C	4.3 min
Depth SNR	6.1
Residual scatter	0.66 %

Light curve

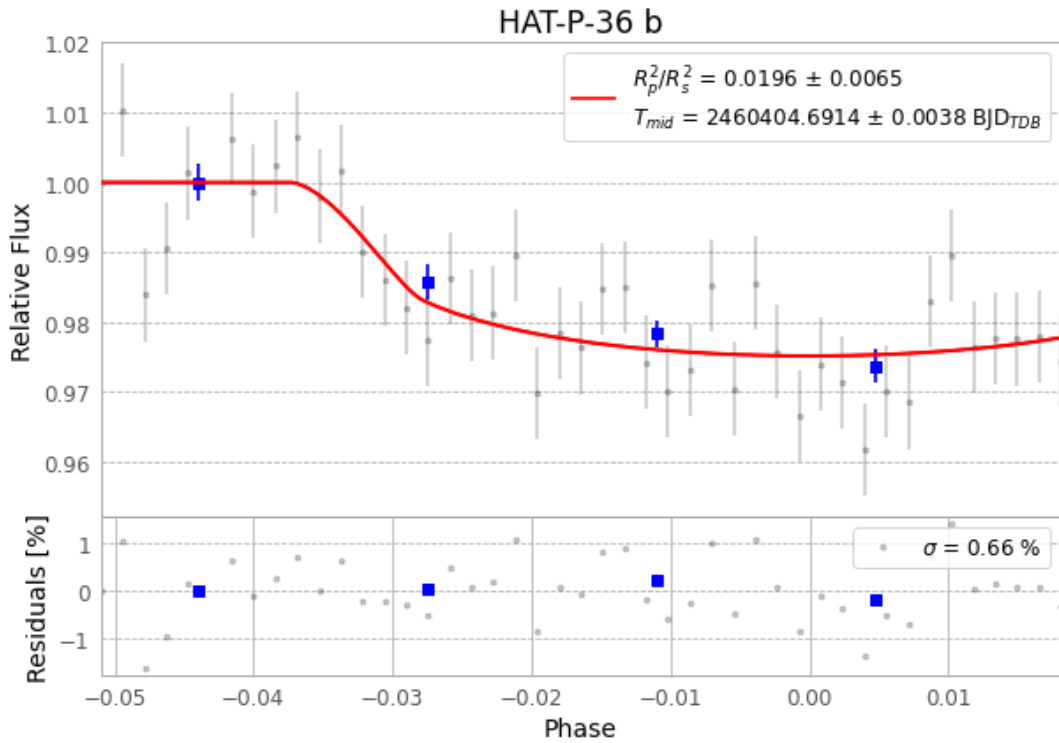


Figure 1 — FinalLightCurve_HAT-P-36 b_2024-04-03.png (SHA-256 79ddfb17776f5e77...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 235], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ff7e072f201e0514...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

45 FITS frames (30 MB), HATP-36240404025115.FITS → HATP-36240404050315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-240404.log (aefc0245ea77...), FinalLightCurve_HAT-P-36 b_2024-04-03.png (79ddfb17776f...), FinalLightCurve_HAT-P-36 b_2024-04-03.csv (61c3235d0234...)

Compute resources

Wall time 110.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 20:56Z.